

Day 4 — Understanding the Benchmark Datasets Used in LLM/KG Research

LLM Research Log | Souvik Pal

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Understanding the Benchmark Datasets Used in LLM/KG Research

Task given: Analyze and understand the benchmark datasets referenced across the papers studied so far — how each is constructed, its working principle, its statistics, and its original source — and record the source link for each dataset. This entry documents each dataset independently, in my own understanding, with the original paper cited as the source of truth.

1. HotpotQA

Task type: Multi-hop Question Answering

Source: Yang et al., "HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering," EMNLP 2018 (ACL Anthology D18-1259, arXiv:1809.09600)

Construction: Crowdworkers were shown pairs of related Wikipedia articles and asked to write questions that genuinely require information from both to answer, rather than being answerable from either one alone. This produced roughly 113,000 question-answer pairs.

Working principle: Each question requires reading and reasoning over multiple supporting documents rather than a single passage. What makes it distinctive is that it also provides sentence-level supporting facts for every answer — so a model isn't just scored on the final answer, but can be evaluated on whether it retrieved the correct evidence to reach it. It also contains "comparison" questions (e.g., comparing two entities' attributes), testing a reasoning skill distinct from plain fact chaining.

Known limitation: Later research (2WikiMultiHopQA) found that a meaningful fraction of HotpotQA's "multi-hop" questions can be answered via shortcuts — surface-level cues — without a model actually performing multi-hop reasoning.

Why it matters for our research: It directly tests whether a model can chain together separate pieces of information — the core capacity that "structured reasoning" approaches (Days 1–3) are trying to strengthen.

2. 2WikiMultiHopQA

Task type: Multi-hop Question Answering with explicit reasoning paths

Source: Ho et al., "Constructing A Multi-hop QA Dataset for Comprehensive Evaluation of Reasoning Steps," COLING 2020 (ACL Anthology 2020.coling-main.580, arXiv:2011.01060)

Construction: Built by combining structured data from Wikidata with unstructured Wikipedia text, specifically to correct a flaw the authors identified in earlier multi-hop datasets: many "multi-hop" questions could secretly be answered without doing multi-hop reasoning at all.

Working principle: Every question is paired with an explicit evidence/reasoning path — the exact chain of facts required to reach the answer — so a model's intermediate reasoning steps, not just its final answer, can be verified. This closes the shortcut-answering gap left open by HotpotQA.

Why it matters for our research: It's a stricter, more reliable benchmark for confirming that a model is actually reasoning step-by-step, rather than pattern matching to a shortcut — directly relevant to validating multi-hop claims made in Day 1's frameworks.

3. ComplexWebQuestions (CWQ)

Task type: Complex, compositional Question Answering over the web/KB

Source: Talmor & Berant, "The Web as a Knowledge-Base for Answering Complex Questions," NAACL 2018 (ACL Anthology N18-1059, arXiv:1803.06643)

Construction: Built on top of an existing KBQA dataset by algorithmically combining simple questions using logical/compositional operations (conjunctions, superlatives, comparisons), then having crowdworkers rephrase them into natural, fluent questions.

Working principle: The authors' key idea is that a complex question can often be decomposed into a sequence of simpler questions, each answerable independently, with the final answer computed by chaining the results together. Their experiments showed this decomposition strategy improved precision@1 from 20.8% (answering the question directly) to 27.5% — a meaningful gain from breaking the question apart rather than treating it as one opaque query.

Why it matters for our research: It's a strong benchmark for testing whether structured or KG-based methods can break down compositional reasoning the way this decomposition strategy does — relevant to evaluating both KG-Adapter (Day 2) and structured reasoning approaches (Day 3).

4. WebQuestionsSP (WebQSP)

Task type: Knowledge Base Question Answering (KBQA)

Source: Yih et al., "The Value of Semantic Parse Labeling for Knowledge Base Question Answering," ACL 2016 (ACL Anthology P16-2033)

Construction: Extends the earlier WebQuestions dataset (questions originally collected via the Google Suggest API and answered by crowdworkers, Berant et al. 2013) by having annotators write full semantic parses — structured SPARQL queries — for each question, grounded against the Freebase knowledge base. This resulted in 4,737 fully-annotated questions, split roughly 3,098 for training and 1,639 for testing.

Working principle: Because each question has a gold-standard structured query attached, models can be trained and evaluated not just on whether they produce the right final answer, but on whether they correctly parse natural language into a structured KB query. The original paper found this extra

annotation effort paid off directly: adding semantic parse labels improved a state-of-the-art KBQA system's F1 score by a full 5 points.

Why it matters for our research: It's one of the standard benchmarks used to evaluate KG-Adapter-style methods (Day 2), since it directly tests grounding language into structured KB queries — exactly the mechanism KG-Adapter is designed to perform.

5. GrailQA

Task type: Knowledge Base Question Answering — generalization testing

Source: Gu et al., "Beyond I.I.D.: Three Levels of Generalization for Question Answering on Knowledge Bases," WWW/TheWebConf 2021 (arXiv:2011.07743)

Construction: Contains 64,331 questions spanning 3,720 relations, 1,442 entity types, and 86 domains from Freebase — deliberately built to be far broader in domain coverage than prior KBQA datasets.

Working principle: The authors argue that standard "train and test on similar data" (i.i.d.) evaluation is misleading for KBQA, since real-world questions can never be fully anticipated in advance. GrailQA instead evaluates models at three explicit levels of generalization: i.i.d. (similar to training data), compositional (new combinations of known elements), and zero-shot (entirely unseen relations/entity types).

Why it matters for our research: This is a much harder and more realistic test of whether a KG-integrated model (like KG-Adapter, Day 2) genuinely generalizes, rather than memorizing surface patterns from its training distribution.

6. MetaQA

Task type: Multi-hop KBQA with noisy/paraphrased questions

Source: Zhang et al., "Variational Reasoning for Question Answering with Knowledge Graph," AAAI 2018 (arXiv:1709.04071)

Construction: Built over a movie-domain knowledge graph (extending the WikiMovies dataset), with over 400,000 questions spanning 1-, 2-, and 3-hop reasoning, plus paraphrased text variants and, in some released versions, spoken/audio question variants — deliberately introducing the kind of noise real users produce.

Working principle: MetaQA addresses two practical problems simultaneously: (1) real user questions contain typos, rephrasing, and pronunciation variation, which makes entity matching against the KG hard, and (2) many of those same questions require multi-hop reasoning to answer. The authors propose a variational deep learning approach that handles noisy input while learning multi-hop reasoning at the same time.

Why it matters for our research: It highlights a real-world constraint often missed in clean academic benchmarks — that a research method also needs to be robust to imperfect, noisy user input, not just clean textbook-style questions.

7. T-REx

Task type: Knowledge Graph ↔ Natural Language alignment resource (not QA)

Source: Elsahar et al., "T-REx: A Large Scale Alignment of Natural Language with Knowledge Base Triples," LREC 2018 (ACL Anthology L18-1544)

Construction: Built by automatically aligning Wikipedia abstracts with Wikidata knowledge base triples using a combination of rule-based and deep learning alignment techniques. The final resource contains exactly 11 million triples aligned to 3.09 million Wikipedia abstracts (6.2 million sentences) — at the time of publication, two orders of magnitude larger than any prior alignment dataset.

Working principle: Unlike the QA datasets above, T-REx is training infrastructure: it doesn't test a model's answer, it teaches a model the correspondence between free-form language and structured facts. This is the dataset used in the Day 3 paper (Zhang et al.) as the training source for knowledge-graph-infused fine-tuning.

Why it matters for our research: It's foundational — this is exactly the kind of resource needed to teach a model how free text maps onto structured graph facts, which underpins the Structured Reasoning approach in Day 3.

8. FEVER

Task type: Fact Extraction and Verification

Source: Thorne et al., "FEVER: a Large-scale Dataset for Fact Extraction and Verification," NAACL 2018 (ACL Anthology N18-1074, arXiv:1803.05355)

Construction: Annotators generated over 185,000 claims by altering sentences taken from Wikipedia, then a separate, independent group of annotators verified each claim against Wikipedia evidence without knowledge of which sentence the claim originally came from — reducing bias in the labeling process.

Working principle: Each claim is labeled Supported, Refuted, or NotEnoughInfo, with the exact supporting or refuting evidence sentence(s) recorded alongside it. Notably, the original paper found this task genuinely difficult — even when given the correct evidence, their best full pipeline reached only 31.87% accuracy, and dropped further, to 50.91%, if evidence was ignored entirely — showing that fact verification against text is far from a solved problem.

Why it matters for our research: This is the dataset behind the fact-verification loss term I'm proposing in Day 3 — it's specifically built to test whether a claim is actually supported by evidence, which is precisely the gap in hallucination measurement I'm trying to close.

9. TruthfulQA

Task type: Truthfulness / Hallucination benchmark

Source: Lin et al., "TruthfulQA: Measuring How Models Mimic Human Falsehoods," ACL 2022 (ACL Anthology 2022.acl-long.229, arXiv:2109.07958)

Construction: The authors deliberately crafted 817 questions across 38 categories (health, law, finance, politics, conspiracy theories, etc.), specifically targeting topics where a model that simply imitates common human text is likely to reproduce a popular misconception rather than the truth.

Working principle: Models are scored on whether their answers are truthful, using either human evaluation or a fine-tuned "truth classifier" model. The paper's most striking finding: their best model was truthful only about 58% of the time, compared to roughly 94% for human performance — and, notably, larger models tended to be less truthful. As a concrete example from the paper, the 6-billion-parameter GPT-J model was 17% less truthful than its own much smaller 125-million-parameter counterpart, since bigger models are better at fluently reproducing convincing-sounding misconceptions absorbed from their training data.

Why it matters for our research: This result is central to my own research motivation — it's direct empirical evidence that scaling model size alone does not fix hallucination, which is exactly why structural/architectural interventions (Days 1–3), rather than scale alone, are necessary.

Cross-Dataset Analysis

Looking at all nine datasets together, I noticed they fall into three distinct functional categories, which matters for how I plan to use them:

- **Training/alignment resources** (T-REx) — teach a model the correspondence between text and structured facts; used during training, not for evaluation.
- **Reasoning benchmarks** (HotpotQA, 2WikiMultiHopQA, ComplexWebQuestions, WebQuestionsSP, GrailQA, MetaQA) — test whether a model can correctly retrieve and chain facts, with increasing levels of strictness (from HotpotQA's basic multi-hop check, up to GrailQA's zero-shot generalization test).
- **Truthfulness/verification benchmarks** (TruthfulQA, FEVER) — test whether a model's output is actually true, independent of whether the reasoning steps look correct. This category is the one the original Day 3 paper (Zhang et al.) never used, and is the specific gap my research proposal targets.

I also noticed a pattern across the reasoning benchmarks: several of them (2WikiMultiHopQA, GrailQA) exist specifically because earlier datasets (HotpotQA, standard KBQA sets) had hidden weaknesses — shortcut-answerable questions, or i.i.d. train/test bias — that made reported results look better than the models' true reasoning ability. This tells me benchmark design itself is an active, evolving research problem, not a fixed, settled foundation.

Summary Table

Dataset	Type	Size (verified)	Primary Use in My Research
HotpotQA	Multi-hop QA	113,000 QA pairs	Multi-hop reasoning evaluation

Dataset	Type	Size (verified)	Primary Use in My Research
2WikiMultiHop QA	Multi-hop QA + reasoning path	~192,600 examples	Verifying genuine reasoning, not shortcuts
ComplexWeb Questions	Compositional QA	—	Testing question decomposition (20.8%→27.5% precision@1)
WebQuestion sSP	KBQA (Freebase)	4,737 questions	Semantic parsing benchmark
GrailQA	KBQA generalization	64,331 questions	i.i.d./compositional/zero-shot generalization
MetaQA	Multi-hop KBQA (noisy)	400,000+ questions	Robustness to real-world noisy input
T-REx	KG–text alignment	11M triples / 3.09M abstracts	Training data for KG-infused fine-tuning
FEVER	Fact verification	185,445 claims	Fact-verification loss term (my proposal)
TruthfulQA	Truthfulness benchmark	817 questions, 38 categories	Core hallucination-rate evaluation

What I Learned

- Not all "datasets" serve the same purpose — some are training resources (T-REx), some are reasoning benchmarks (HotpotQA, 2WikiMultiHopQA), and some are evaluation-only truthfulness tests (TruthfulQA, FEVER). Conflating these categories would make a research design incoherent.
- Benchmark design itself is an active research problem — GrailQA and 2WikiMultiHopQA both exist specifically because earlier datasets had hidden flaws (shortcut answers, i.i.d. assumptions) that made results look better than they really were.
- The TruthfulQA finding that larger models can be less truthful is a particularly important piece of evidence for my research direction: it justifies why my proposal focuses on structural fixes rather than assuming scale will eventually solve hallucination on its own.

Personal Reflection

Going through nine datasets in one sitting was a lot, but it helped me see how they connect — like some datasets were made just to fix problems in older ones. There's a lot to know about this, and I'm still learning how all the pieces fit together.